

United States Top 50 Music Analytics and Playlist Performance Analysis

A Data Analytics Research Paper

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Course

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An exploratory research study of playlist rankings, song longevity, popularity trends, artist dominance, content attributes, and performance KPIs using a United States Top 50 music dataset.

Prepared using the structure and analytical framing of the supplied reference research paper, adapted to the actual findings documented in Project 2.ipynb.

Abstract

This research paper investigates performance patterns within a United States Top 50 playlist using a structured music dataset containing song, artist, popularity, ranking, duration, album type, album size, explicit-content status, and album-cover information. The study follows the organization of the supplied reference research paper—introduction, problem definition, literature review, methodology, results, discussion, recommendations, limitations, conclusion, references, and appendices—while adapting advanced predictive sections to the descriptive and exploratory analytics actually performed in Project 2.ipynb.

The original project dataset contains 22,994 rows and 10 columns. During preprocessing, 96 duplicate song-date observations were identified on 01-03-2025 and duplicate date-position records were removed, producing 22,944 rows at that stage. The project subsequently generated a cleaned/exported analytical dataset reported as 27,750 rows and 19 columns. Because this final shape differs from the initial source shape, the discrepancy is explicitly documented as a data-pipeline limitation rather than silently reconciled. The final analysis identifies 861 unique songs and 273 unique artists.

The results show strong variation in chart longevity, artist presence, ranking movement, and popularity. “Something in the Orange” has the longest recorded playlist presence at 536 days, while “un x100to” has the highest average popularity score among songs meeting the project’s seven-day minimum at 99.20. Taylor Swift has the greatest playlist breadth with 96 unique songs and the highest Artist Dominance Index at 7.26%. Popularity and rank are only weakly correlated ($r = -0.094$), while best rank and chart longevity show a moderate relationship ($r = -0.342$). Explicit and non-explicit songs have very similar performance, whereas singles outperform album and compilation tracks on average popularity and average rank.

Keywords: Music Analytics, Playlist Analytics, Spotify, Top 50, Song Popularity, Rank Volatility, Artist Dominance, Chart Longevity, Exploratory Data Analysis, Data Analyst

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1. Introduction

Digital music platforms generate high-frequency observations of song position, artist presence, popularity, and content characteristics. A Top 50 playlist is particularly suitable for analytics because the same songs can be observed repeatedly across daily snapshots, allowing the analyst to study not only where a song ranks but also how long it remains visible and how its ranking changes over time.

Project 2.ipynb frames the analysis around ranking stability and volatility, song longevity and popularity trends, dominant artists and songs, explicit versus non-explicit content, singles versus album tracks, duration and album-size effects, and the development of key performance indicators. These objectives create a complete descriptive analytics workflow from raw playlist observations to interpretable performance measures.

The supplied reference paper emphasizes that business analytics should progress from data acquisition and preparation through EDA, feature engineering, evaluation, interpretation, recommendations, limitations, and future research. This paper adopts that structure. However, because Project 2 does not document supervised prediction, SHAP explanations, or formal clustering, those components are treated as methodological extensions rather than presented as completed results.

2. Problem Definition

The central problem is to determine what characteristics are associated with strong and sustained playlist performance in the United States Top 50. Raw rank alone is insufficient: a song can reach a high position briefly, remain mid-chart for a long period, or fluctuate considerably while maintaining popularity. The analysis therefore combines ranking, longevity, movement, popularity, artist presence, content attributes, and album structure.

A second problem concerns interpretation. Popularity scores, playlist ranks, and artist dominance measure different aspects of performance. A robust analyst must therefore avoid treating any one metric as a complete definition of success. Project 2 addresses this by creating song-level KPIs and comparing multiple dimensions.

The practical decision question is: **Which measurable song, artist, ranking, and content characteristics are most useful for understanding sustained Top 50 playlist performance?**

3. Objectives and Research Questions

Objectives

- Analyze ranking stability and volatility.
- Measure song longevity and popularity trends.
- Identify dominant artists and high-performing songs.
- Compare explicit and non-explicit content performance.
- Compare singles, album tracks, and compilation tracks.
- Study the relationship of song duration and album size with popularity and rank.
- Construct song-level and portfolio-level KPIs.
- Translate the findings into practical recommendations for playlist and content analytics.

Research Questions

1. How stable is the Top 50 ranking structure and how frequently do songs move up or down?
2. Which songs show the strongest chart longevity and average popularity?
3. Do strong peak ranks correspond to longer playlist presence?
4. Which artists dominate playlist breadth, days of presence, and total appearances?
5. How strongly does popularity explain playlist position?

6. Does explicit status meaningfully affect performance?
7. Do singles outperform album and compilation tracks?
8. Do duration and album size materially influence popularity or rank?

4. Literature Review

The literature review is adapted to the music-analytics setting while preserving the reference paper's purpose: establish the conceptual foundation for the analytical choices and clarify how descriptive evidence can support decision-making.

4.1 Music and Digital-Platform Analytics

Digital music analytics commonly relies on repeated observations of content performance, audience signals, rankings, and creator-level activity. In this project, playlist position is treated as an observable platform-performance signal, while popularity provides a complementary content-level metric. The repeated daily structure enables longitudinal descriptive analysis.

4.2 Exploratory Data Analysis

EDA is appropriate because the first requirement is to understand distributions, missing values, duplicates, rank coverage, artist concentration, and relationships among numerical variables. Project 2 uses inspection, aggregation, correlations, bar charts, histograms, scatter plots, and grouped comparisons before interpreting performance.

4.3 Ranking, Popularity and Longevity

Playlist rank and popularity are related but not identical. A ranking represents relative placement within the Top 50 at a particular snapshot, whereas popularity is a score attached to the track. Longevity measures repeated playlist presence. The project therefore evaluates them separately and then examines their correlations.

4.4 Segmentation and Artist Dominance

The reference paper uses customer segmentation to transform model outputs into interpretable personas. Project 2 adapts this idea to artist and song-level descriptive segments: high-breadth artists, high-longevity songs, high-popularity songs, and high-volatility songs. These are rule-based analytical groups, not formal clustering models.

4.5 Explainability and Predictive Analytics: Methodological Position

The reference paper contains logistic regression, random forest, XGBoost, SHAP, model-performance metrics, back-testing, propensity tiers, and k-means segmentation. Project 2 does not document these procedures. Accordingly, this paper does not fabricate AUC, precision, recall, SHAP, cluster scores, or model predictions. These techniques are presented only as future extensions where appropriate.

4.6 Research Gap

The project-level gap is the need for an integrated, interpretable view of playlist performance. Simple rank tables do not reveal whether a song's success is durable, whether popularity explains ranking, which artists dominate, or whether content attributes matter. The project addresses this gap through multi-dimensional KPIs and comparative EDA.

5. Methodology

The methodology follows the reference paper's sequence of acquisition, data description, quality preparation, EDA, feature engineering, evaluation, results interpretation, recommendations, and limitations. The predictive/XAI elements of the reference are intentionally marked as not performed in this project.

5.1 Data Acquisition

The source file used in Project 2 is `Atlantic_United_States.csv`. The initial load reports 22,994 observations and 10 columns. The variables are `date`, `position`, `song`, `artist`, `popularity`, `duration_ms`, `album_type`, `total_tracks`, `is_explicit`, and `album_cover_url`.

5.2 Data Description

Variable	Type / Meaning
date	Playlist snapshot date
position	Top 50 playlist rank (1–50)
song	Track title
artist	Artist / credited artist field
popularity	Popularity score
duration_ms	Track duration in milliseconds
album_type	Single, album, or compilation
total_tracks	Number of tracks in the associated album
is_explicit	Explicit-content indicator
album_cover_url	Album-cover reference URL

5.3 Data Inspection and Quality Checks

Check	Result
Original shape	22,994 × 10
Initial missingness	1 missing value in each of the 9 non-date fields
Duplicate song-date observations	96
Duplicate date	01-03-2025
Rows on duplicate date	100
Rows after date-position deduplication	22,944
Rank minimum / maximum	1 / 50
Invalid ranks	0
Unique artists after standardization	273
Unique songs in KPI table	861

The project also identifies two date-count anomalies during daily validation: a malformed date value “23-08-” and 43 records on 23-08-2025. The original notebook output therefore demonstrates the importance of explicit date-quality controls.

5.4 Data Cleaning and Pre-processing

- Inspect data types and missing values.
- Identify duplicate song-date records.
- Remove duplicate date-position playlist entries.
- Validate rank boundaries between 1 and 50.
- Convert date to datetime.
- Standardize artist names by trimming whitespace and collapsing repeated spaces.
- Create a duration-in-minutes feature.

A pipeline discrepancy is important: the final notebook output reports a cleaned dataset of 27,750 rows and 19 columns, which is larger than the original 22,994-row source. The paper reports this exactly as documented and treats the change as a reproducibility limitation requiring pipeline review before production use.

5.5 Feature Engineering

- `duration_min` = `duration_ms` / 60,000.
- `days_on_chart` = number of unique dates for each song-artist pair.
- `average_rank` = mean playlist position.
- `best_rank` = minimum playlist position.
- `rank_volatility` = standard deviation of position.
- `previous_rank` and `rank_movement` = previous position minus current position.
- `daily_rank_movement` = movement restricted to consecutive-day appearances.
- `popularity_trend` = seven-day rolling mean popularity.
- `average_popularity` and `popularity_trend_score` at song level.
- `rank_group` = Top 10, Rank 11–20, Rank 21–50.

These features create a compact KPI layer analogous to the reference paper's feature-engineering stage, but they remain interpretable descriptive measures.

5.6 Analytical Metrics

Metric	Definition	Purpose
Days on Chart	Unique observed dates per song	Longevity
Average Rank	Mean position	Typical chart performance
Best Rank	Minimum position	Peak performance
Rank Volatility	Standard deviation of position	Stability / movement
Rank Movement	Previous rank – current rank	Direction of change
Popularity Trend	Seven-day rolling mean	Short-term popularity stability
Artist Dominance Index	Artist appearances / total observations × 100	Artist share of playlist presence
Average Popularity	Mean popularity by song/group	Popularity comparison

5.7 Evaluation Strategy

The reference paper uses predictive evaluation, back-testing, precision-oriented metrics, calibration, cumulative gains, and SHAP. Project 2 instead uses descriptive evaluation: rank distributions, frequency counts, group means, correlations, longevity measures, dominance shares, and comparative tables. Correlation coefficients are interpreted as associations, not causal effects.

5.8 Ethical and Practical Considerations

The dataset contains song and artist metadata rather than personal learner or customer information. Album-cover URLs are not used as a personal profiling mechanism. Any production dashboard should respect licensing and platform terms for artwork and metadata, and should clearly distinguish descriptive analytics from causal claims.

6. Results

The results chapter follows the reference paper's progression from data quality and EDA to segmentation-like interpretation, KPI evaluation, and decision implications.

6.1 Data Quality and Distribution Analysis

The initial dataset contains 22,994 rows and 10 columns. One row has missing values across the nine non-date fields, and 96 duplicate song-date observations were identified. Duplicate date-position records were removed, leaving 22,944 rows. Rank validation found no values outside 1–50.

The final exported cleaned dataset is reported by the notebook as 27,750 rows and 19 columns. Since this cannot be obtained simply by removing duplicates from the original file, the difference should be investigated before treating the exported file as a direct one-to-one transformation of the source.

6.2 Playlist Ranking Analysis

The rank distribution covers positions 1 through 50, consistent with the intended Top 50 structure. Rank movement analysis shows 11,374 downward movements, 8,889 upward movements, and 1,819 observations with no change among consecutive appearances. Downward movement is therefore more frequent than upward movement in the recorded transitions.

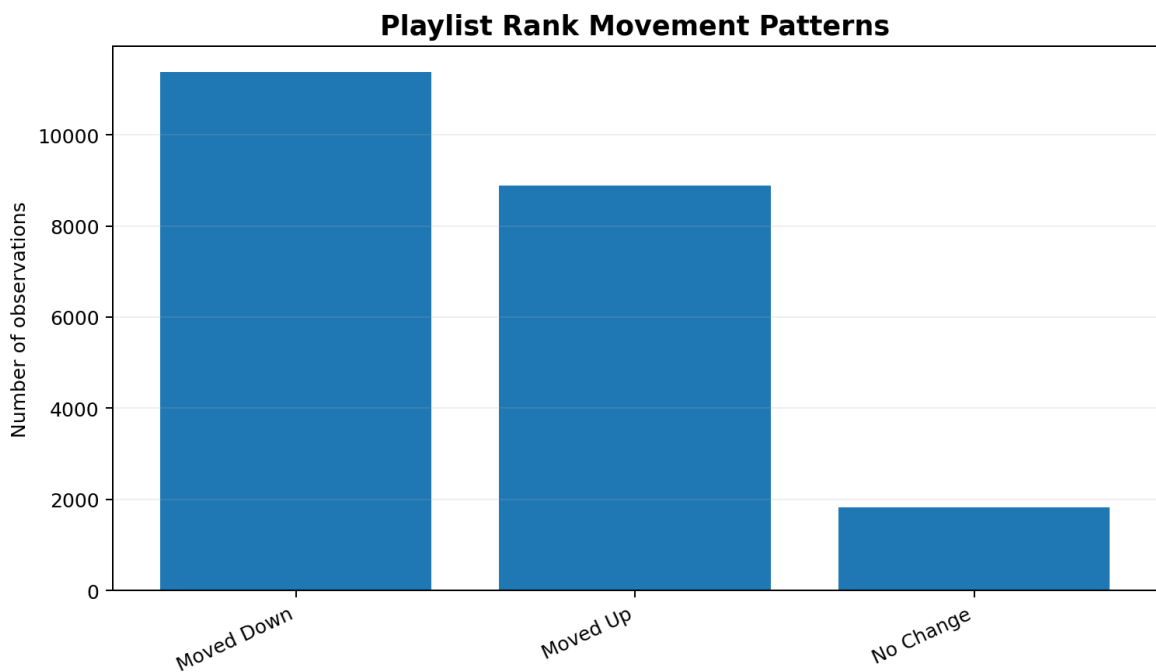


Figure 1. Playlist rank movement patterns.

Movement Type	Observations	Share of classified movements
Moved Down	11,374	51.58%
Moved Up	8,889	40.30%
No Change	1,819	8.24%
Total	22,082	100.00%

6.3 Rank Movement and Stability

The project constructs previous_rank and rank_movement and then creates a corrected daily movement measure restricted to consecutive-day appearances. The largest volatility examples include “Cruel Summer - Live from TS | The Eras Tour” (25.46), “Stories About My Brother” (24.75), and “FUK SUMN” (23.37). These high-volatility tracks illustrate that peak position and stability are distinct performance dimensions.

Song	Artist	Rank Volatility
Cruel Summer - Live from TS The Eras Tour	Taylor Swift	25.46
Stories About My Brother	Drake	24.75
FUK SUMN	Kanye West & Ty Dolla \$ign	23.37
As Good As I Once Was	Toby Keith	22.63
Wildflower and Barley	Hozier	22.63

6.4 Song-Level Performance

The song-level KPI table contains 861 unique song-artist combinations. It combines longevity, average rank, best rank, rank volatility, average popularity, and popularity trend.

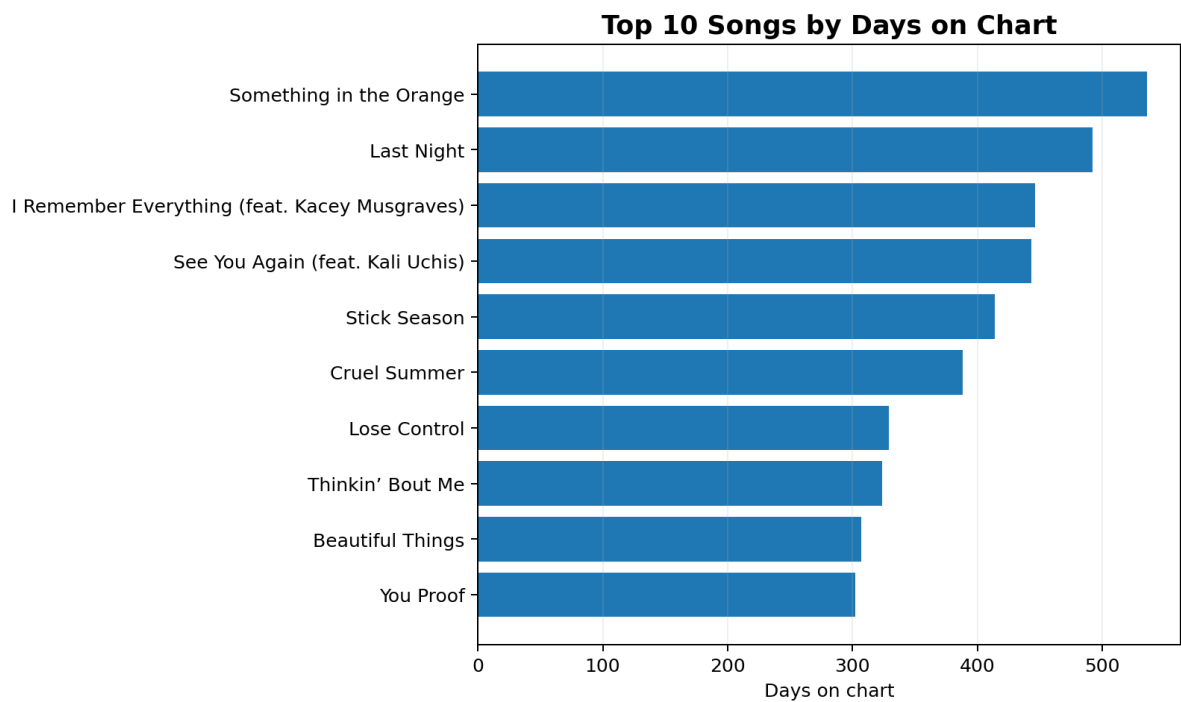


Figure 2. Top 10 songs by days on chart.

“Something in the Orange” records the longest playlist presence at 536 days, followed by “Last Night” at 492 days. Long-lived songs represent sustained playlist relevance rather than merely one-time peak performance.

Song	Artist	Days on Chart
Something in the Orange	Zach Bryan	536
Last Night	Morgan Wallen	492
I Remember Everything (feat. Kacey Musgraves)	Zach Bryan	446
See You Again (feat. Kali Uchis)	Tyler, The Creator	443
Stick Season	Noah Kahan	414
Cruel Summer	Taylor Swift	388
Lose Control	Teddy Swims	329
Thinkin' Bout Me	Morgan Wallen	324
Beautiful Things	Benson Boone	307
You Proof	Morgan Wallen	302

6.5 Peak Rank and Longevity

The correlation between best rank and days on chart is -0.342. Because lower numerical rank is better, the negative coefficient indicates that stronger peak positions tend to accompany longer chart presence. However, the relationship is moderate rather than deterministic: a strong peak does not guarantee long-term persistence.

6.6 Popularity Performance

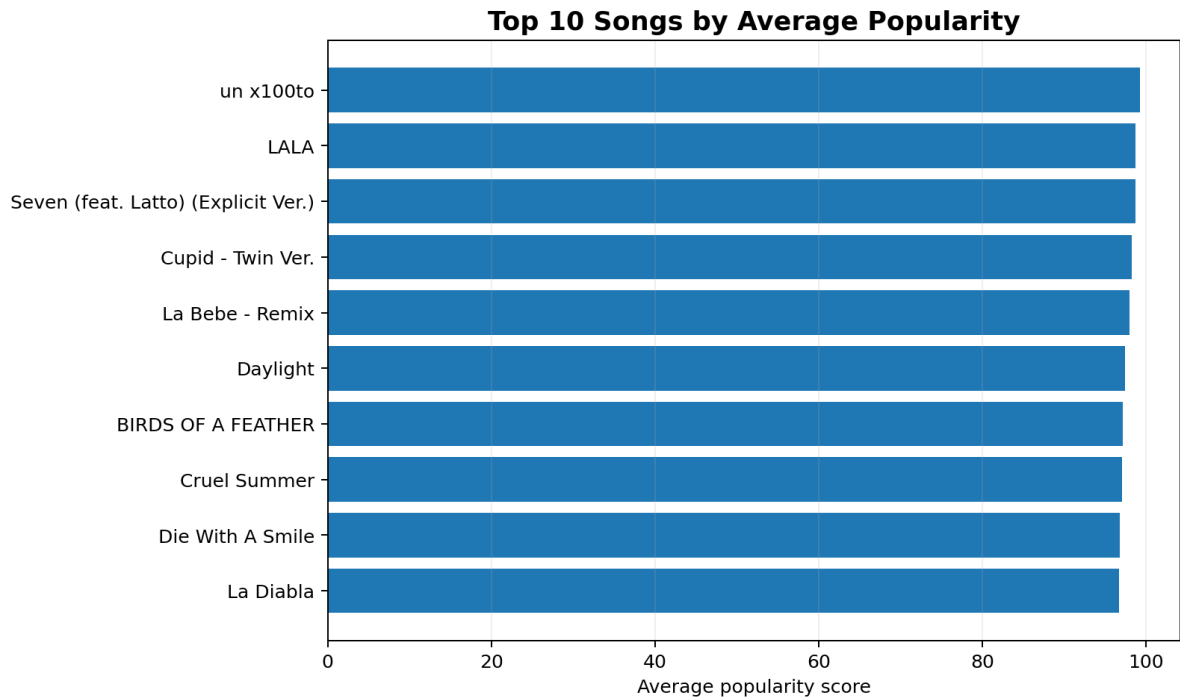


Figure 3. Top 10 songs by average popularity among songs with at least seven observed chart days.

“un x100to” has the highest average popularity score at 99.20, followed by LALA at 98.69 and Seven (feat. Latto) at 98.65. “Cruel Summer” is notable because it combines very high popularity (97.01) with 388 days on chart.

Song	Artist	Avg. Popularity	Days
un x100to	Grupo Frontera & Bad Bunny	99.2	79
LALA	Myke Towers	98.69	49
Seven (feat. Latto) (Explicit Ver.)	Jung Kook & Latto	98.65	81
Cupid - Twin Ver.	FIFTY FIFTY	98.2	71
La Bebe - Remix	Yng Lvcas & Peso Pluma	98.01	113
Daylight	David Kushner	97.4	55
BIRDS OF A FEATHER	Billie Eilish	97.16	191
Cruel Summer	Taylor Swift	97.01	388
Die With A Smile	Lady Gaga & Bruno Mars	96.81	103
La Diabla	Xavi	96.67	104

6.7 Average Rank and Peak Performance

Song	Artist	Average Rank	Best Rank
MELTDOWN	Travis Scott	1.0	1.0
I Had Some Help (Feat. Morgan Wallen)	Post Malone	2.14	1.0
FE!N	Travis Scott	2.33	2.0
Not Like Us	Kendrick Lamar	2.81	1.0
Please Please Please	Sabrina Carpenter	3.57	1.0
I Had Some Help (Feat. Morgan Wallen)	Post Malone & Morgan Wallen	3.91	1.0

Song	Artist	Average Rank	Best Rank
Seven (feat. Latto)	Jung Kook & Latto & BTS	4.0	4.0
Parade on Cleveland (feat. Drake)	Young Thug	4.0	4.0
BIRDS OF A FEATHER	Billie Eilish	4.46	2.0
Who	Jimin	4.48	1.0

MELTDOWN has the strongest average rank at 1.00, followed by I Had Some Help (2.14) and FE!N (2.33). Several songs reach rank 1, demonstrating that a track can have a strong peak while its average performance depends on how long it remains at high positions.

6.8 Artist Performance

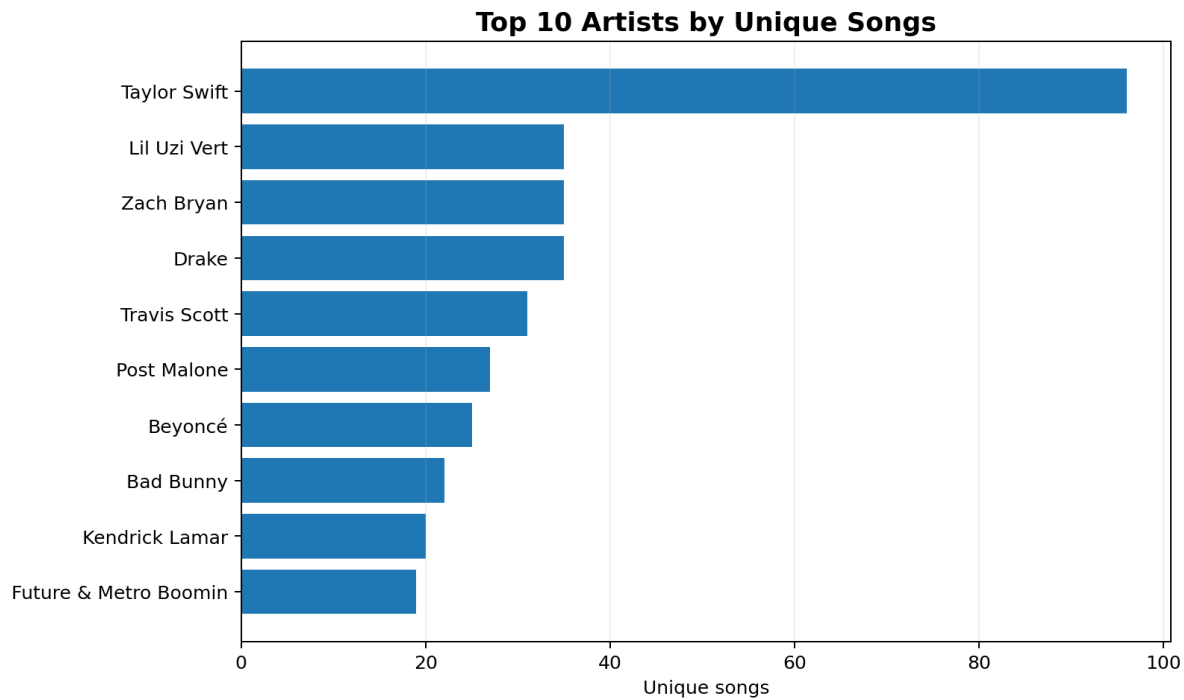


Figure 4. Top 10 artists by unique songs on the playlist.

Taylor Swift dominates playlist breadth with 96 unique songs, substantially exceeding the next artists at 35 each. This indicates exceptionally broad representation across the observed playlist records.

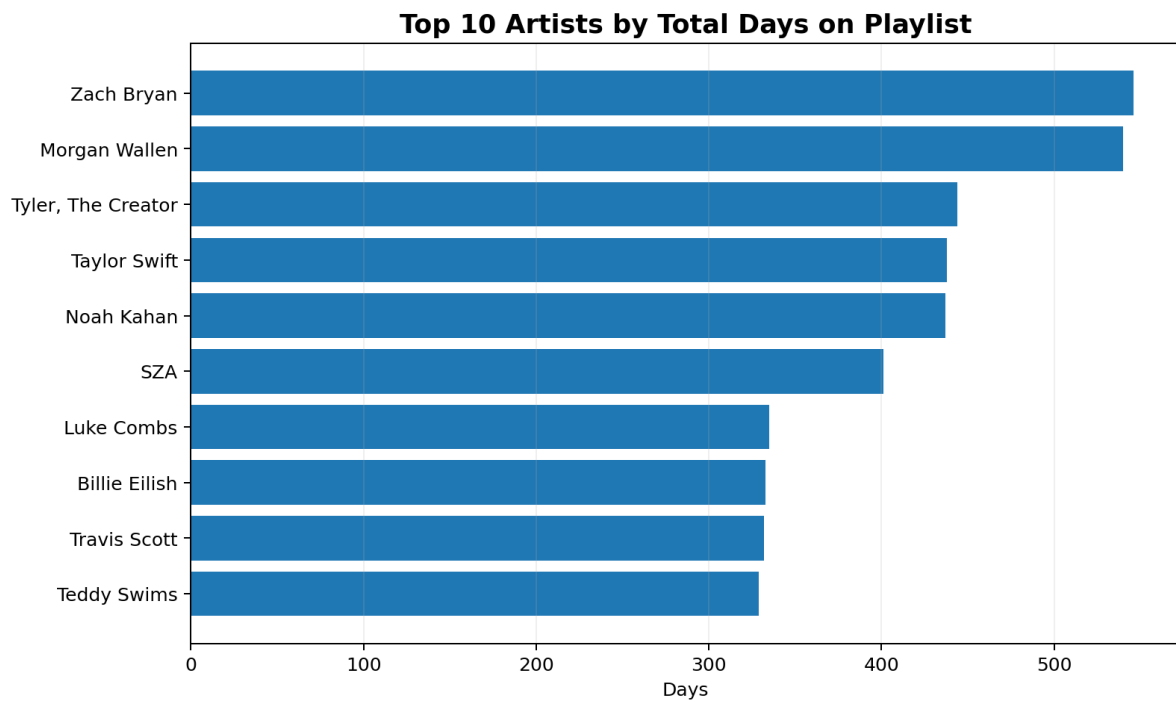


Figure 5. Top 10 artists by total days of playlist presence.

Zach Bryan leads total artist presence with 546 days, followed by Morgan Wallen with 540 and Tyler, The Creator with 444. Breadth and sustained presence are therefore different dimensions of artist success.

6.9 Artist Dominance Index

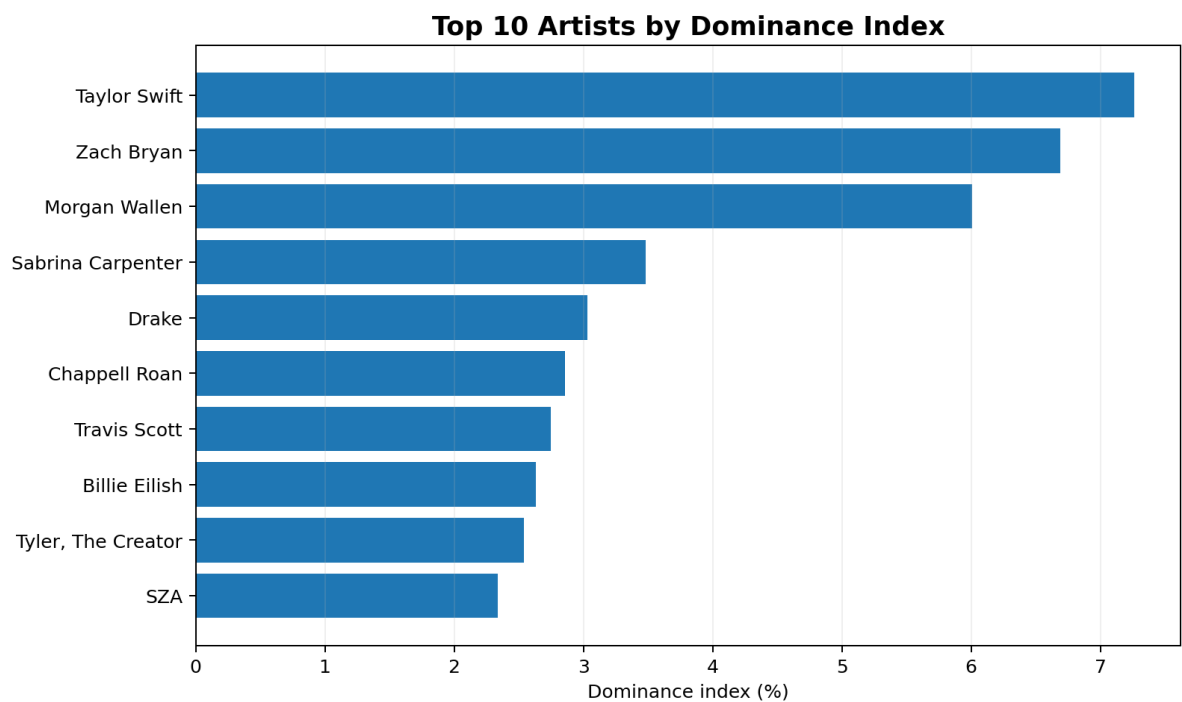


Figure 6. Top 10 artists by Artist Dominance Index.

Taylor Swift has the highest Artist Dominance Index at 7.26%, followed by Zach Bryan at 6.69% and Morgan Wallen at 6.01%. The index is based on total playlist appearances and therefore captures repeated presence rather than unique-song breadth alone.

Artist	Total Appearances	Dominance Index
Taylor Swift	2015	7.26
Zach Bryan	1857	6.69
Morgan Wallen	1669	6.01
Sabrina Carpenter	967	3.48
Drake	841	3.03
Chappell Roan	794	2.86
Travis Scott	762	2.75
Billie Eilish	729	2.63
Tyler, The Creator	705	2.54
SZA	648	2.34

The project also finds that artist dominance changes over time, with Taylor Swift showing strong monthly peaks while Zach Bryan, Drake, Morgan Wallen, and Sabrina Carpenter lead during different periods. This temporal variation supports the use of trend-based artist monitoring rather than a static leaderboard.

6.10 Popularity Score Analytics

Playlist rank and popularity have a very weak negative correlation of -0.094. This means popularity alone does not strongly explain playlist position.

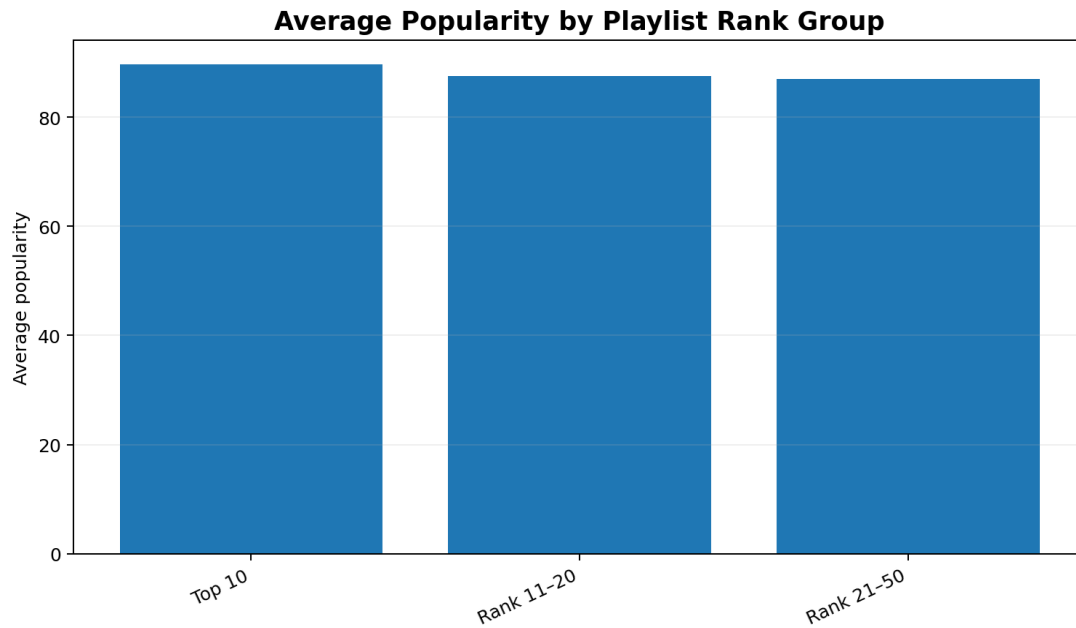


Figure 7. Average popularity by playlist rank group.

Rank Group	Average Popularity
Top 10	89.61
Rank 11–20	87.59
Rank 21–50	87.03

Top 10 songs have the highest average popularity (89.61), but the difference from ranks 11–20 (87.59) and ranks 21–50 (87.03) is modest. The project also reports a weak positive correlation of 0.16 between popularity volatility and rank volatility, suggesting limited linkage between popularity changes and chart instability.

6.11 Content Attribute Analysis

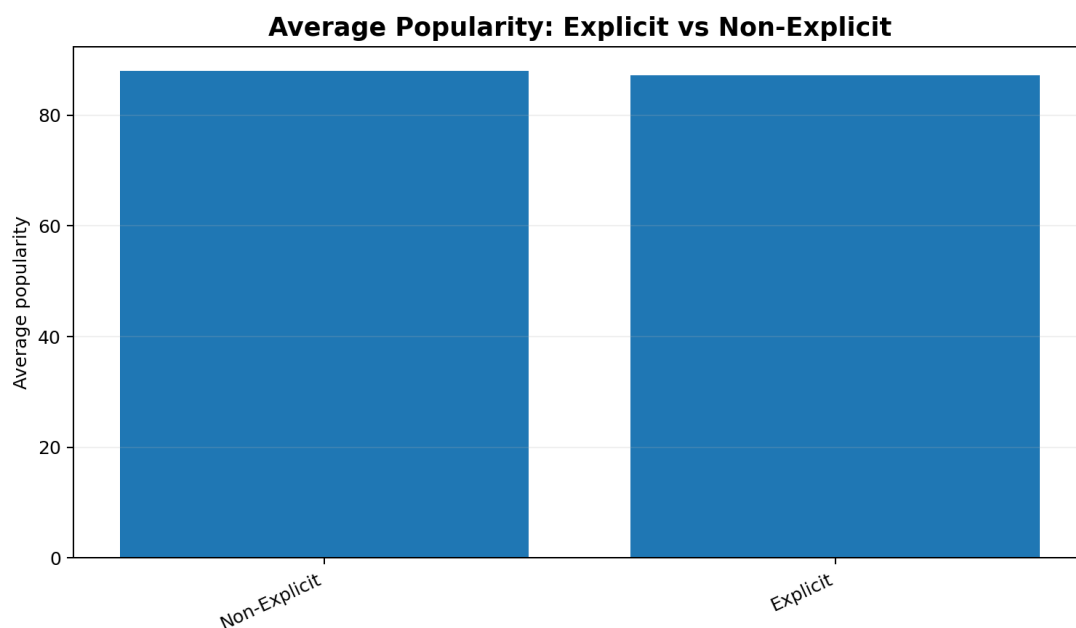


Figure 8. Average popularity by explicit-content status.

Content Type	Average Popularity	Average Rank
Non-Explicit	87.99	25.6
Explicit		

Content Type	Average Popularity	Average Rank
Explicit	87.3	25.39

Explicit and non-explicit songs perform similarly: non-explicit tracks average 87.99 popularity and rank 25.60, while explicit tracks average 87.30 popularity and rank 25.39. Explicit content therefore shows little overall association with playlist performance. Explicit content represents 47.94% of the observed records, while non-explicit content represents 52.06%.

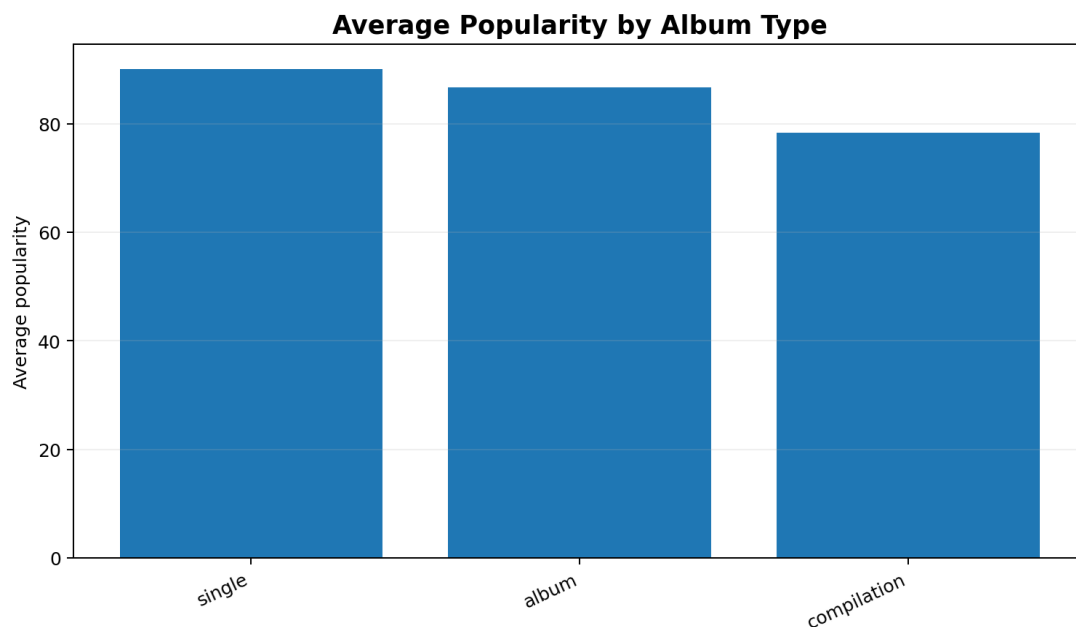


Figure 9. Average popularity by album type.

Album Type	Average Popularity	Average Rank	Unique Songs
single	90.13	23.99	239
album	86.68	26.1	745
compilation	78.38	33.19	8

Singles outperform album and compilation tracks on average popularity, with a score of 90.13 and average rank 23.99. Albums average 86.68 popularity and rank 26.10, while the small compilation group averages 78.38 and rank 33.19. The small number of unique compilation songs means that this comparison should be interpreted cautiously.

6.12 Duration and Album Size

Relationship	Correlation
Duration vs Popularity	-0.043
Duration vs Rank	-0.045
Album Size vs Popularity	-0.182
Album Size vs Rank	0.116

Song duration has almost no relationship with popularity or playlist rank. Album size shows only weak relationships: larger albums are associated with slightly lower popularity and slightly weaker rankings, but the coefficients are not strong enough to support a major performance effect.

6.13 KPI Summary

KPI	Result
Average days on chart	28.40
Overall average rank	30.05
Average rank volatility index	7.71
Average popularity trend score	75.66
Top artist dominance	Taylor Swift — 7.26%
Explicit content share	47.94%

The KPI layer provides a concise portfolio-level monitoring framework. Average days on chart captures persistence; average rank captures typical placement; volatility measures instability; popularity trend measures recent popularity; dominance measures artist concentration; and explicit share describes content composition.

7. Discussion

The discussion follows the reference paper's principle of translating analytical results into operational and strategic meaning. Because this is a music playlist project, the focus is on playlist monitoring, content strategy, artist visibility, and dashboard decision support rather than banking operations.

7.1 Interpretation of Ranking Behavior

Downward movements outnumber upward movements, indicating that chart observations contain substantial competitive displacement. The existence of 1,819 no-change observations also shows that ranking persistence is common enough to warrant stability metrics. A playlist manager should therefore monitor both direction and magnitude of change.

7.2 Song Longevity and Popularity

Longevity and popularity identify different kinds of success. "un x100to" leads average popularity among the selected songs, whereas "Something in the Orange" leads longevity. "Cruel Summer" demonstrates that high popularity can coexist with exceptional persistence. These patterns support multi-KPI evaluation instead of a single ranking score.

7.3 Artist Dominance

Taylor Swift leads unique-song breadth and total appearances, while Zach Bryan leads total days of presence. This distinction matters: an artist can dominate through many different tracks or through unusually persistent tracks. Artist reporting should therefore include unique songs, days present, total appearances, and dominance index.

7.4 Content and Album-Type Effects

Explicit status shows minimal difference in overall performance. Album type is more differentiated, with singles showing stronger average popularity and rank. However, descriptive comparisons cannot establish that release format causes performance; factors such as release timing, artist popularity, promotion, and song age may confound the relationship.

7.5 Operational Dashboard Implications

The analysis maps naturally into a dashboard containing a playlist timeline explorer, song ranking trends, artist dominance leaderboard, popularity-versus-rank analysis, explicit-content comparison, album-type comparison, and filters for date, artist, song, rank range, and album type. These views allow decision-makers to move from portfolio-level KPIs to individual-song investigation.

8. Recommendations

- 1. Use a multi-KPI definition of success.** Combine average rank, best rank, days on chart, popularity, volatility, and trend rather than relying on one measure.
- 2. Monitor longevity separately from popularity.** High popularity does not automatically imply the longest playlist life.
- 3. Track artist breadth and persistence separately.** Report unique songs, total days, total appearances, and dominance index.
- 4. Prioritize trend monitoring.** Weekly or monthly movement views can reveal emerging artists and declining presence earlier than static rankings.
- 5. Use rank-volatility alerts.** Large changes can identify songs requiring deeper review of release timing, promotion, or competitive effects.
- 6. Maintain explicit-content neutrality in performance interpretation.** The observed averages are very similar, so explicit status should not be treated as a strong performance predictor without additional evidence.
- 7. Evaluate singles and album tracks with contextual controls.** The current descriptive result favors singles, but future analysis should control for release date, artist, and song age.

8. Improve the data pipeline before production deployment. Resolve the discrepancy between the 22,994-row source and the reported 27,750-row final export and validate malformed/missing dates.

9. Extend the dashboard with cohort and release-age views. Compare new releases with older tracks and study how long songs take to reach peak rank.

10. Use controlled experiments for causal decisions. If playlist placement or promotional actions are changed, evaluate them through controlled tests rather than inferring causality from correlations.

9. Limitations

- The project is descriptive; correlations do not establish causal effects.
- The final exported dataset shape ($27,750 \times 19$) differs from the original loaded shape ($22,994 \times 10$), and the exact transformation causing the increase is not documented in the notebook outputs.
- Two daily-count anomalies are present, including a malformed date and a day with fewer than 50 records.
- Album-type comparisons are uneven because compilation contains only eight unique songs.
- Popularity is treated as an observed score; the project does not model how the score itself is generated.
- The analysis does not include streaming counts, listener demographics, playlist editorial decisions, marketing exposure, release dates, or external social-media signals.
- Artist names may combine collaborations into a single artist string, which can affect artist-level dominance measures.
- No supervised ML, SHAP, formal clustering, ROC/AUC, precision/recall, calibration, or causal/uplift modeling is performed.
- Album-cover URLs are metadata references and are not used as analytical evidence.

10. Conclusion

This study provides a structured Data Analyst view of United States Top 50 playlist performance. Following the supplied reference research paper's comprehensive organization, it connects data acquisition, quality checks, preprocessing, feature engineering, EDA, performance measurement, segmentation-like interpretation, discussion, recommendations, limitations, and future work.

The findings show that chart performance is multi-dimensional. "Something in the Orange" has the longest recorded playlist presence at 536 days, while "un x100to" has the highest average popularity score among songs observed for at least seven days. Taylor Swift has the largest unique-song breadth at 96 and the highest Artist Dominance Index at 7.26%. Popularity has only a weak relationship with rank ($r = -0.094$), whereas best rank has a moderate relationship with longevity ($r = -0.342$). Explicit and non-explicit tracks perform similarly, while singles show the strongest average popularity and rank.

The project therefore demonstrates that a playlist should not be evaluated through rank alone. A combined framework of longevity, average rank, peak rank, volatility, popularity trend, artist dominance, and content attributes provides a richer decision-support system. The resulting KPIs and visualizations are directly suitable for an interactive Streamlit music analytics dashboard.

Most importantly, the study maintains methodological discipline: advanced methods from the reference paper are not presented as completed analyses when they are absent from Project 2. Instead, they are retained as future extensions. This makes the paper comprehensive while remaining faithful to the evidence contained in the actual project notebook.

11. Future Work

- Build song-level predictive models for future rank movement or chart survival.
- Use survival analysis to estimate the probability that a song remains in the Top 50.
- Apply clustering to create formal song or artist performance personas.

- Use SHAP or other explainability methods if a validated predictive model is developed.
- Add release-date and song-age features to distinguish new-release effects from enduring popularity.
- Model artist-level concentration over time and identify periods of rapid dominance change.
- Develop cohort analysis for songs entering the Top 50 during different periods.
- Test whether playlist position changes have measurable downstream effects on popularity using controlled or quasi-experimental designs.
- Automate data-quality validation for duplicate positions, malformed dates, missing fields, and daily 50-entry coverage.
- Extend the Streamlit dashboard with alerts, downloadable KPI summaries, and interactive song/artist drill-downs.

12. References

The supplied reference research paper is used primarily for research-paper organization and methodological framing. General analytical references are listed where they support the concepts of exploratory analysis, machine learning, segmentation, and interpretability.

- [1] James, G., Witten, D., Hastie, T., Tibshirani, R., & Taylor, J. (2023). *An Introduction to Statistical Learning: With Applications in Python*. Springer.
- [2] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794.
- [3] Reutterer, T., & Dan, D. (2019). *Cluster Analysis in Marketing Research*.
- [4] Bücker, M., Szepannek, G., Gosiewska, A., & Biecek, P. (2020). Transparency, Auditability and Explainability of Machine Learning Models in Credit Scoring. *arXiv:2009.13384*.
- [5] Davenport, T., Guha, A., Grewal, D., & Bressgott, T. (2020). How Artificial Intelligence Will Change the Future of Marketing. *Journal of the Academy of Marketing Science*, 48(1), 24–42.
- [6] Kannan, P. K., & Li, H. (2017). Digital Marketing: A Framework, Review and Research Agenda. *International Journal of Research in Marketing*, 34(1), 22–45.
- [7] Lessmann, S., Baesens, B., Seow, H.-V., & Thomas, L. C. (2015). Benchmarking State-of-the-Art Classification Algorithms for Credit Scoring. *European Journal of Operational Research*, 247(1), 124–136.
- [8] Moro, S., Cortez, P., & Rita, P. (2014). A Data-Driven Approach to Predict the Success of Bank Telemarketing. *Decision Support Systems*, 62, 22–31.
- [9] Mühlhling, P. M. (2025). Using Machine Learning to Address Real Banking Challenges: Profiling Customer Behaviour and Campaign Fatigue. Supplied reference research paper used for structural and methodological framing.
- [10] Project 2.ipynb. United States Top 50 Music Analytics project notebook supplied by the author. Source of dataset preparation, feature engineering, analysis outputs, charts, and KPI results.

Appendix A. Analytical Tables

A1. Core Dataset and Quality Summary

Item	Result
Original observations	22,994
Original columns	10
Rows after duplicate date-position removal	22,944
Final exported rows reported by notebook	27,750
Final exported columns reported by notebook	19
Unique songs	861
Unique artists	273
Duplicate song-date observations identified	96
Invalid rank values	0

A2. Rank Movement

Movement	Count
Moved Down	11374
Moved Up	8889
No Change	1819
Total classified transitions	22,082

A3. Explicit Content

Content	Share	Average Popularity	Average Rank
Non-Explicit	52.06%	87.99	25.60
Explicit	47.94%	87.30	25.39

A4. Album Type

Type	Average Popularity	Average Rank	Unique Songs
single	90.13	23.99	239
album	86.68	26.1	745
compilation	78.38	33.19	8

Appendix B. Figure Index

Figure 1. Playlist rank movement patterns.

Figure 2. Top 10 songs by days on chart.

Figure 3. Top 10 songs by average popularity.

Figure 4. Top 10 artists by unique songs.

Figure 5. Top 10 artists by total days of playlist presence.

Figure 6. Top 10 artists by Artist Dominance Index.

Figure 7. Average popularity by playlist rank group.

Figure 8. Average popularity by explicit-content status.

Figure 9. Average popularity by album type.

Appendix C. Project Traceability

Reference-paper component	Implementation in this music-analytics paper
Cover / Abstract / Keywords	Title, author, Data Analyst course, project abstract and keywords
Introduction	Music-platform context and analytical motivation
Problem Definition	Multi-dimensional playlist-performance problem
Literature Review	EDA, ranking, popularity, segmentation, interpretability
Data Acquisition / Description	Source dataset, columns, dimensions, analytical roles
Cleaning / Pre-processing	Missing values, duplicates, rank checks, dates, artist standardization
EDA	Rank distribution, movement, longevity, popularity, artist and content comparisons
Feature Engineering	Duration, longevity, rank metrics, movement, popularity trend, rank groups
Performance Metrics	Descriptive KPI framework instead of predictive metrics
Segmentation	Descriptive song/artist performance groups; no unsupported clustering
Modeling / XAI	Explicitly not performed; retained as future extension
Results	Song, artist, popularity, content, album, duration and album-size findings
Discussion	Interpretation and operational dashboard implications
Recommendations	Playlist monitoring, artist strategy, data-quality and future analytics
Limitations	Pipeline discrepancy, anomalies, missing contextual variables, descriptive scope
Conclusion	Integrated summary of evidence and analytical contribution
Future Work	Prediction, survival analysis, clustering, SHAP, cohort and causal testing
References / Appendices	Analytical references, tables, figures and traceability

Source-integrity note: The paper deliberately does not copy the reference paper's banking-specific findings into the music project. Its structure and analytical concepts are adapted, while all numerical music findings are taken from Project 2.ipynb outputs.